








Time Series Analysis

Seminar 6: VAR models and Granger causality



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CCF**: Cross-Correlation Function
- **EODHD**: EOD Historical Data (market-data provider)
- **FEVD**: Forecast Error Variance Decomposition
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **HQ**: Hannan–Quinn (information criterion)
- **OLS**: Ordinary Least Squares
- **ROBOR**: Romanian Interbank Offer Rate
- **SE**: Standard Error
- **TSA**: Time Series Analysis
- **VAR**: Vector AutoRegression

Today's Question and Seminar Structure

- The question of the seminar
  - **when we model several series together, which of them helps to forecast the others, and how does a shock in one spread to the rest?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	a VAR(1) by hand: stability, mean, forecasts	Solved, Proposed	A1
A3, A4	orthogonalised impulse responses; a variance decomposition	Solved, Proposed	A3
A5, A6	a Granger F test; lag selection by AIC, BIC and HQ	Solved, Proposed	A5
B1, B2	S&P 500 and BET: daily; weekly	Solved, Proposed	B1
B3, B4	a Romanian macro VAR; the EUR/RON rate added	Solved, Proposed	B3
C1, C2	the price puzzle; what is wrong in an AI answer?	Proposed	B3

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** open questions, an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
S&P 500, BET	EODHD	daily, close	2000–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, seasonally adjusted	2005–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2004–2026
ROBOR 3M	Eurostat (irt_st_m)	monthly, % per year	2005–2026
EUR/RON	BNR reference rate	daily	2005–2026

- Quarterly VAR variables: $g_t = 100\Delta \ln \mathrm{GDP}_t$; π_t: 12-month HICP inflation (quarterly mean); i_t: ROBOR 3M (quarterly mean); s_t: $100\Delta \ln$ of the quarterly mean EUR/RON
- In the notebook: `VAR` from `statsmodels`, `granger_F`, `ic_table`, `girf`; no account or key is needed

Prerequisites for Today (1/17): the VAR model

- **VAR**(p) Sims (1980): each variable depends on the last p values of all variables

$$\mathbf{Y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{Y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t, \qquad E[\boldsymbol{\varepsilon}_t] = \mathbf{0}, \quad E[\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t'] = \boldsymbol{\Sigma}$$

  - $\mathbf{Y}_t$: the vector of the K series at date t; $\mathbf{c}$: the $K \times 1$ vector of constants; $\mathbf{A}_j$: $K \times K$ coefficient matrices
  - $\boldsymbol{\varepsilon}_t$: white-noise shocks, uncorrelated over time; $\boldsymbol{\Sigma}$: their covariance matrix, elements σ_{jk}; $'$: the transpose

- $K = 2$, $p = 1$, equation by equation

$$y_{1t} = c_1 + a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + \varepsilon_{1t}, \qquad y_{2t} = c_2 + a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + \varepsilon_{2t}$$

  - a_{jk}: the effect of $y_{k,t-1}$ on y_{jt}; $a_{12} \neq 0$: the past of y_2 helps to predict y_1
  - $\sigma_{12} \neq 0$: the two shocks of the same period are correlated

Prerequisites for Today (2/17): stability and eigenvalues

- **Stability** of a VAR(1): all eigenvalues λ of $\mathbf{A}$ satisfy $|\lambda| < 1$

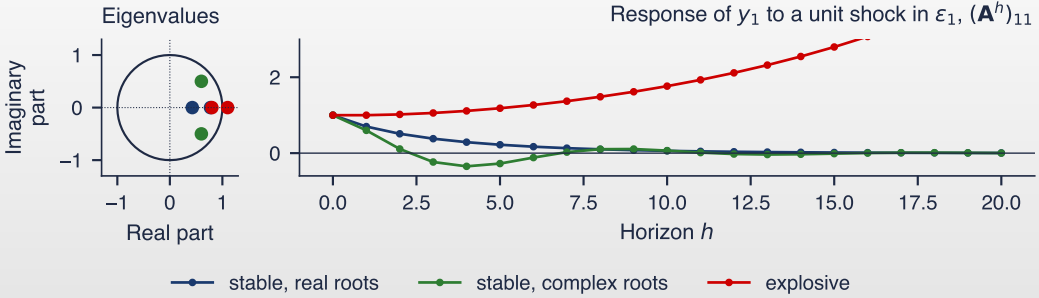
$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \overset{K=2}{\iff} \lambda^2 - \operatorname{tr}(\mathbf{A})\,\lambda + \det(\mathbf{A}) = 0$$

  - $\mathbf{I}$: the identity matrix; $\operatorname{tr}(\mathbf{A}) = a_{11} + a_{22}$: the trace; $\det(\mathbf{A}) = a_{11}a_{22} - a_{12}a_{21}$: the determinant
  - complex roots $\lambda = a \pm bi$: modulus $|\lambda| = \sqrt{a^2 + b^2}$; the responses oscillate
  - stable: the effect of a shock dies out at the rate $\max|\lambda|$ per period; $|\lambda| > 1$: explosive
- **VAR**(p): the eigenvalues of the companion matrix (here $p = 2$)

$$\begin{pmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{Y}_{t-1} \\ \mathbf{Y}_{t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ \mathbf{0} \end{pmatrix}$$

  - a VAR(p) written as a VAR(1) of dimension Kp; stable if all Kp eigenvalues have modulus below 1

Prerequisites for Today (3/17): stable, oscillating and explosive VARs



- Left: the eigenvalues of three 2×2 matrices and the unit circle; right: element $(1, 1)$ of $\mathbf{A}^h$
- Moduli: 0.77 and 0.43 (real roots, smooth decay); 0.78 twice (complex roots, damped oscillation)
  - 1.10 and 0.80: one root outside the circle, the response grows without limit

TSA_ch6_seminar

Prerequisites for Today (4/17): mean, forecasts and estimation

- **Mean** of a stable VAR(1): the level to which the series return

$$\boldsymbol{\mu} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{c}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

  - from $\boldsymbol{\mu} = \mathbf{c} + \mathbf{A}\boldsymbol{\mu}$, taking expectations in the model; a, b, c, d: the elements of a 2×2 matrix

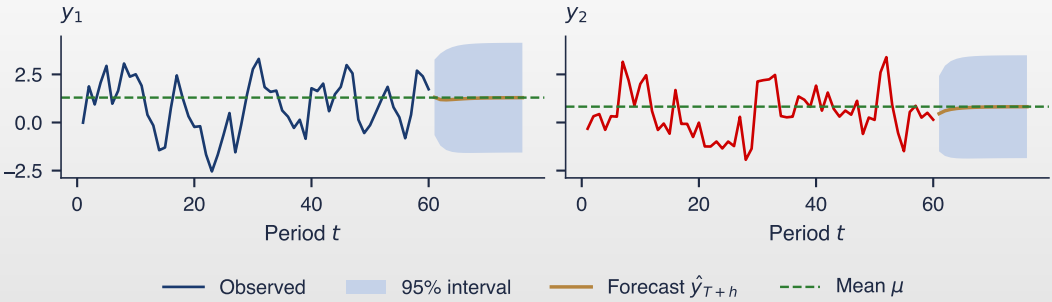
- **Forecasts** by recursion, with the future shocks replaced by 0

$$\hat{\mathbf{Y}}_{T+1} = \mathbf{c} + \mathbf{A}\mathbf{Y}_T, \qquad \hat{\mathbf{Y}}_{T+h} = \mathbf{c} + \mathbf{A}\hat{\mathbf{Y}}_{T+h-1} \longrightarrow \boldsymbol{\mu}$$

  - $\mathbf{Y}_T$: the last observation; error covariance $\boldsymbol{\Sigma}_h = \sum_{s=0}^{h-1} \mathbf{A}^s \boldsymbol{\Sigma} (\mathbf{A}^s)'$, 95% interval $\pm 1.96\sqrt{(\boldsymbol{\Sigma}_h)_{jj}}$

- **Estimation**: OLS equation by equation; all equations have the same regressors
  - $K + pK^2$ coefficients; the variables must be stationary (Chapter 3): growth rates, returns, rates without trend

Prerequisites for Today (5/17): a simulated VAR(1) and its forecasts



- $\mathbf{A} = \begin{pmatrix} 0.5 & 0.3 \\ 0.1 & 0.6 \end{pmatrix}$, $\mathbf{c} = (0.4, 0.2)'$, $\sigma_{11} = \sigma_{22} = 1$, $\sigma_{12} = 0.5$: eigenvalues 0.73 and 0.37, mean $\boldsymbol{\mu} = (1.29, 0.82)'$
- The forecasts move from $\mathbf{Y}_T = (1.73, 0.14)'$ towards $\boldsymbol{\mu}$
  - the interval widens with h: standard deviation 1.00 at $h = 1$, 1.46 and 1.36 at $h = 16$ (the unconditional values)

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Prerequisites for Today (6/17): the lag order

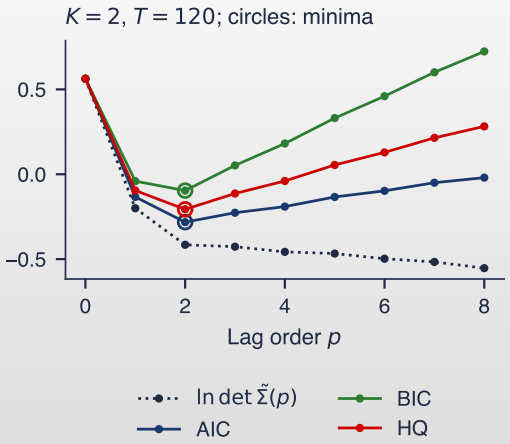
- **Information criteria**: fit against the number of coefficients; the smallest value wins

$$\mathrm{IC}(p) = \ln\det\tilde{\mathbf{\Sigma}}(p) + c_T\,\frac{pK^2}{T}, \qquad \tilde{\mathbf{\Sigma}}(p) = \frac{1}{T}\sum_t \hat{\varepsilon}_t\hat{\varepsilon}_t'$$

  - $\hat{\varepsilon}_t$: the residuals of the VAR(p); T: the observations, on a common sample for all p; pK^2: the lag coefficients
  - $\ln\det\tilde{\mathbf{\Sigma}}(p)$ always falls with p; the penalty $c_T\,pK^2/T$ grows with p

- **Penalty per coefficient** c_T
  - $c_T = 2$: AIC Akaike (1974); $c_T = \ln T$: BIC Schwarz (1978); $c_T = 2\ln\ln T$: HQ Hannan and Quinn (1979)
  - for $T \geq 16$: $2 < 2\ln\ln T < \ln T$, so BIC chooses the smallest p, AIC the largest, HQ in between

Prerequisites for Today (7/17): information criteria on simulated data



- A simulated VAR(2), $K = 2$
  - VARs with $p = 0, \ldots, 8$ fitted on the same $T = 120$ observations
- $\ln\det\tilde{\boldsymbol{\Sigma}}(p)$ keeps falling
  - extra lags always fit the sample better
- All three criteria have their minimum at $p = 2$
  - after it, BIC rises fastest

Prerequisites for Today (8/17): residual diagnostics

- **Portmanteau test** Hosking (1980): no auto- or cross-correlation of the residuals up to lag h

$$Q_h = T \sum_{j=1}^{h} \operatorname{tr}(\hat{\mathbf{C}}_j' \hat{\mathbf{C}}_0^{-1} \hat{\mathbf{C}}_j \hat{\mathbf{C}}_0^{-1}) \sim \chi^2(K^2(h-p)), \qquad \hat{\mathbf{C}}_j = \frac{1}{T} \sum_{t=j+1}^{T} \hat{\boldsymbol{\varepsilon}}_t \hat{\boldsymbol{\varepsilon}}_{t-j}'$$

  - $\hat{\mathbf{C}}_j$: the residual autocovariance matrix at lag j; the multivariate version of Ljung–Box
  - H_0: white-noise residuals; a small p-value: add lags or variables

- **Jarque–Bera** Jarque and Bera (1980): normality of the residuals of one equation

$$\mathrm{JB} = \frac{T}{6}\left[S^2 + \frac{(\kappa-3)^2}{4}\right] \sim \chi^2(2)$$

  - S: the skewness, κ: the kurtosis of the residuals; often rejected because of crises, which does not invalidate OLS

Prerequisites for Today (9/17): cross-correlations

- **Cross-correlation** at lag k: does x of k periods ago move with y today?

$$\hat{\rho}_{yx}(k) = \widehat{\text{Corr}}(y_t, x_{t-k}) = \frac{\sum_t (y_t - \bar{y})(x_{t-k} - \bar{x})}{\sqrt{\sum_t (y_t - \bar{y})^2 \sum_t (x_t - \bar{x})^2}}$$

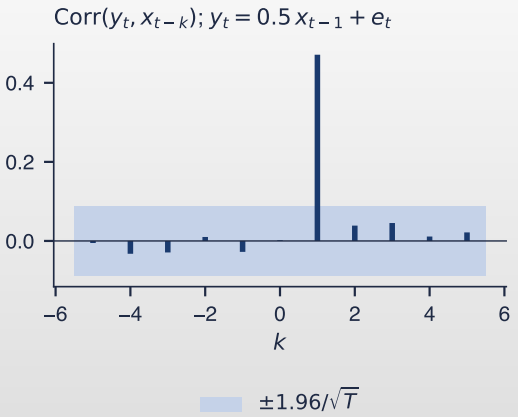
  - $k > 0$: x leads y; $k < 0$: y leads x; $k = 0$: the same period
  - band $\pm 1.96/\sqrt{T}$ if the two series are independent white noise

- **Weekly returns**: the sum of the daily log returns of a week

$$r_w = 100 \ln \frac{P_{\text{Friday}}}{P_{\text{previous Friday}}}$$

  - a lead of one day falls inside the same week: it moves from lag 1 to lag 0

Prerequisites for Today (10/17): a series that leads another



- Simulated: $y_t = 0.5\,x_{t-1} + e_t$
  - x_t, e_t: independent white noise, $T = 500$
- One spike, at $k = 1$: 0.47
  - x leads y by one period
- All other lags inside ± 0.088
  - no same-period correlation ($k = 0$: 0.00)

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Prerequisites for Today (11/17): Granger causality

- x **Granger-causes** y Granger (1969): the past of x improves the forecast of y beyond the past of y and of the other variables
- In the equation of y of a VAR(p)

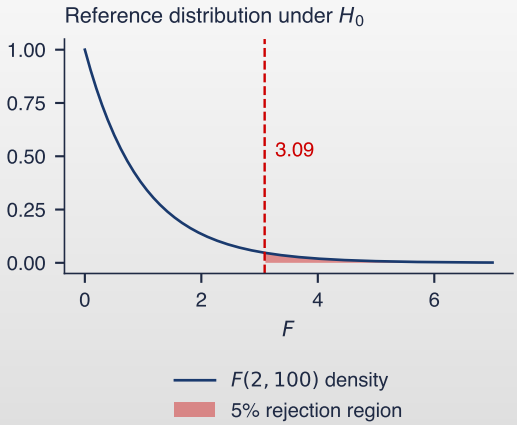
$$y_t = c + \sum_{j=1}^{p} a_j y_{t-j} + \sum_{j=1}^{p} b_j x_{t-j} + e_t, \qquad H_0 \colon b_1 = \cdots = b_p = 0$$

  - a_j, b_j: the coefficients of the lags of y and of x (other variables of the VAR enter as well); e_t: the error
- F **test**: does dropping the lags of x raise the residual sum of squares by much?

$$F = \frac{(RSS_R - RSS_U)/q}{RSS_U/(T-k)} \sim F(q,\ T-k)$$

  - RSS_U, RSS_R: residual sums of squares with and without the lags of x; $q = p$: the restrictions; k: the coefficients of the unrestricted equation

Prerequisites for Today (12/17): the F test and its limits



- Reject H_0 if F exceeds the 95% quantile
  - $F(2, 100)$: 3.09
  - p-value: the area to the right of F
- **Not** cause and effect
  - predictability, relative to the variables chosen
- Pitfalls
  - an omitted common driver
  - effects in the same period: **instantaneous causality**, visible only in $\boldsymbol{\Sigma}$
  - non-stationary series

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Prerequisites for Today (13/17): impulse responses

- **Moving-average form** of a stable VAR(1): the series as a sum of past shocks

$$\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \boldsymbol{\Phi}_h \boldsymbol{\varepsilon}_{t-h}, \qquad \boldsymbol{\Phi}_h = \mathbf{A}^h$$

  - element (j, k) of $\boldsymbol{\Phi}_h$: the response of y_j after h periods to a unit shock in ε_k, all else 0
- **Orthogonalised shocks**: the shocks are correlated, so we write them with uncorrelated ones

$$\boldsymbol{\varepsilon}_t = \mathbf{P}\mathbf{u}_t, \quad \boldsymbol{\Sigma} = \mathbf{P}\mathbf{P}', \qquad \boldsymbol{\Theta}_h = \boldsymbol{\Phi}_h \mathbf{P}$$

  - $\mathbf{u}_t$: uncorrelated shocks with variance 1; $\mathbf{P}$: the lower triangular **Cholesky** factor of $\boldsymbol{\Sigma}$
  - $\theta_{jk,h}$, element of $\boldsymbol{\Theta}_h$: the response of y_j after h periods to a shock u_k of one standard deviation

Prerequisites for Today (14/17): the Cholesky factor and the ordering

- **Cholesky** for $K = 2$

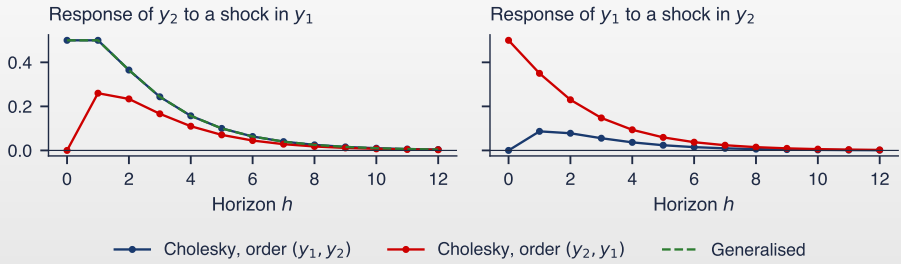
$$\mathbf{P} = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix}, \qquad p_{11} = \sqrt{\sigma_{11}}, \quad p_{21} = \frac{\sigma_{21}}{p_{11}}, \quad p_{22} = \sqrt{\sigma_{22} - p_{21}^2}$$

  - obtained by solving $\mathbf{PP}' = \mathbf{\Sigma}$ element by element
- **The ordering matters**: $p_{12} = 0$, so the first variable does not react in the same period to the shock of the second
  - the correlation of the shocks is attributed entirely to the variable ordered first
- **Generalised responses** Pesaran and Shin (1998): do not depend on the ordering

$$\psi_h^{(k)} = \frac{\mathbf{\Phi}_h \mathbf{\Sigma} \mathbf{e}_k}{\sqrt{\sigma_{kk}}}$$

  - $\mathbf{e}_k$: column k of $\mathbf{I}$; equal to the Cholesky response with variable k ordered first

Prerequisites for Today (15/17): responses under two orderings



- $\mathbf{A} = \begin{pmatrix} 0.5 & 0.1 \\ 0.3 & 0.4 \end{pmatrix}$, $\sigma_{11} = \sigma_{22} = 1$, $\sigma_{12} = 0.5$; responses to shocks of one standard deviation
- On impact: 0.5 for the variable ordered second, 0 for the one ordered first
  - the generalised response of y_2 coincides with the Cholesky response with y_1 first
  - estimated responses come with 95% bands (asymptotic or bootstrap Kilian (1998)); a band containing 0: no evidence of an effect

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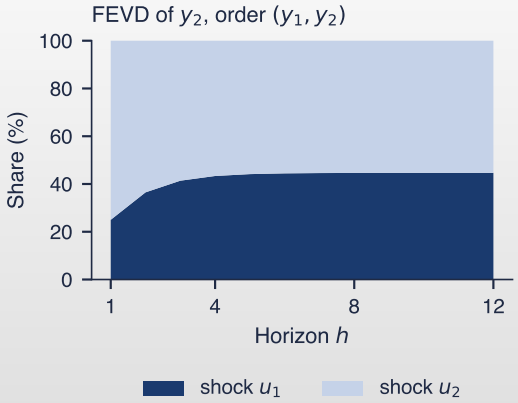
Prerequisites for Today (16/17): variance decomposition

- **FEVD** (forecast error variance decomposition): the share of shock u_k in the h-step forecast error variance of y_j

$$\omega_{jk}(h) = \frac{\sum_{s=0}^{h-1} \theta_{jk,s}^2}{\sum_{s=0}^{h-1} \sum_{m=1}^{K} \theta_{jm,s}^2}$$

  - the numerator: the part of the error variance due to u_k; the denominator: the total h-step error variance of y_j
  - each row sums to 100%; it depends on the ordering, like $\mathbf{\Theta}_h$

Prerequisites for Today (17/17): a variance decomposition



- The VAR of the previous chart
  - order (y_1, y_2)
- $h = 1$: u_1 explains 25% of the variance of y_2
  - $\theta^2_{21,0} = 0.25$ out of $0.25 + 0.75$
- The share grows to 43% at $h = 4$
  - and settles near 45%
  - through $a_{21} = 0.3$

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A1: a VAR(1) by hand [Solved]

Task

- $\mathbf{c} = (0.4, 0.8)'$, $\mathbf{A} = \begin{pmatrix} 0.6 & 0.2 \\ 0.2 & 0.6 \end{pmatrix}$; last observation $\mathbf{Y}_T = (3, 2)'$.
- 1. Write the two equations.
- 2. Find the eigenvalues of $\mathbf{A}$ and decide whether the VAR is stable.
- 3. Compute the mean $\boldsymbol{\mu}$.
- 4. Compute $\hat{\mathbf{Y}}_{T+\mathbf{1}}$ and $\hat{\mathbf{Y}}_{T+\mathbf{2}}$.
- Report: two equations, two eigenvalues, $\boldsymbol{\mu}$, two forecasts.

Solution

- 1. $y_{\mathbf{1}t} = 0.4 + 0.6y_{\mathbf{1},t-\mathbf{1}} + 0.2y_{\mathbf{2},t-\mathbf{1}} + \varepsilon_{\mathbf{1}t}$;
  $y_{\mathbf{2}t} = 0.8 + 0.2y_{\mathbf{1},t-\mathbf{1}} + 0.6y_{\mathbf{2},t-\mathbf{1}} + \varepsilon_{\mathbf{2}t}$
- 2. $\text{tr} = 1.2$, $\det = 0.36 - 0.04 = 0.32$:
  $\lambda^{\mathbf{2}} - 1.2\lambda + 0.32 = 0$, $\lambda = 0.8$ and 0.4; both below 1: stable
- 3. $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 0.4 & -0.2 \\ -0.2 & 0.4 \end{pmatrix}$, $\det = 0.12$;
  $\boldsymbol{\mu} = \frac{\mathbf{1}}{\mathbf{0.12}} \begin{pmatrix} 0.4 & 0.2 \\ 0.2 & 0.4 \end{pmatrix} \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 2.667 \\ 3.333 \end{pmatrix}$
- 4. $\hat{\mathbf{Y}}_{T+\mathbf{1}} = (0.4 + 1.8 + 0.4,\ 0.8 + 0.6 + 1.2)' = \begin{pmatrix} 2.60 \\ 2.60 \end{pmatrix}$;
  $\hat{\mathbf{Y}}_{T+\mathbf{2}} = \begin{pmatrix} 2.48 \\ 2.88 \end{pmatrix}$: moving towards $\boldsymbol{\mu}$

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A2: stable or not? [Proposed]

Solution discussed in class

Task

- Two VAR(1) matrices: (a) $\mathbf{A} = \begin{pmatrix} 0.9 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$; (b) $\mathbf{A} = \begin{pmatrix} 0.5 & -0.4 \\ 0.4 & 0.5 \end{pmatrix}$. Model: A1.
- 1. For each matrix, compute the trace, the determinant and the eigenvalues.
- 2. Decide whether each VAR is stable.
- 3. Describe how the responses to a shock behave in each case.
- Report: four eigenvalues (or moduli), two decisions and two sentences.

A3: orthogonalised impulse responses [Solved]

Task

- The VAR(1) of A1 with $\mathbf{\Sigma} = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}$, order (y_1, y_2).
- 1. Compute the Cholesky factor $\mathbf{P}$.
- 2. Compute $\mathbf{\Theta_0} = \mathbf{P}$ and $\mathbf{\Theta_1} = \mathbf{AP}$, and $\mathbf{\Phi_2} = \mathbf{A^2}$.
- 3. Compute $\mathbf{P}$ again with the order (y_2, y_1) and say what changes on impact.
- Report: three matrices and one sentence.

Solution

- 1. $p_{11} = 1$, $p_{21} = 0.4$, $p_{22} = \sqrt{1 - 0.16} = 0.917$: $\mathbf{P} = \begin{pmatrix} 1.000 & 0.000 \\ 0.400 & 0.917 \end{pmatrix}$
- 2. $\mathbf{\Theta_1} = \begin{pmatrix} 0.680 & 0.183 \\ 0.440 & 0.550 \end{pmatrix}$; $\mathbf{\Phi_2} = \begin{pmatrix} 0.40 & 0.24 \\ 0.24 & 0.40 \end{pmatrix}$; after $u_1 = 1$: y_2 moves by 0.4 at once and by 0.44 one period later
- 3. Order (y_2, y_1): $\mathbf{P} = \begin{pmatrix} 0.917 & 0.400 \\ 0.000 & 1.000 \end{pmatrix}$ (in the original order); now y_1 reacts on impact to the y_2 shock (0.4) and y_2 does not react to the y_1 shock

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A4: a variance decomposition by hand [Proposed]

Solution discussed in class

Task

- Use Θ_0 and Θ_1 of A3 (order y_1, y_2). Model: A3.
- 1. Compute the share of u_1 in the 1-step forecast error variance of y_2.
- 2. Compute the 2-step forecast error variance of y_2 and the share of u_1 in it.
- 3. Compute the share of u_2 in the 2-step error variance of y_1, and explain why it is 0 at one step.
- Report: three percentages and one sentence.

A5: a Granger F test [Solved]

Task

- A bivariate VAR(2) on $T = 100$ observations. In the equation of y_1: $RSS_U = 45.2$ (with the two lags of y_2) and $RSS_R = 52.8$ (without them).
- 1. Give H_0, the number of restrictions q and the number of coefficients k of the unrestricted equation.
- 2. Compute F.
- 3. Decide at 5% and state the conclusion in words.
- Report: H_0, F, a decision and one sentence.

Solution

- 1. H_0: $a_{12}^{(1)} = a_{12}^{(2)} = 0$ (y_2 does not Granger-cause y_1); $q = 2$; $k = 1 + 2 \cdot 2 = 5$
- 2. $F = \dfrac{(52.8 - 45.2)/2}{45.2/95} = \dfrac{3.8}{0.476} = 7.99$
- 3. $7.99 > 3.09$, the 5% value of $F(2, 95)$ ($p < 0.001$): reject; the past of y_2 helps to forecast y_1, which does not prove that y_2 causes y_1

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A6: choosing p by hand [Proposed]

Solution discussed in class

Task

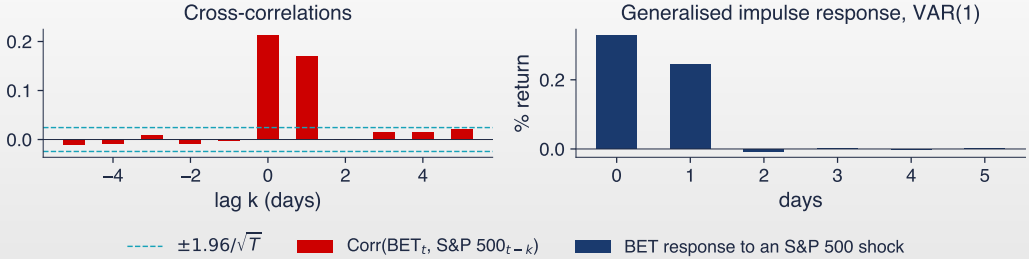
- A VAR with $K = 3$ on a common sample of $T = 80$ quarters gives $\ln\det\tilde{\boldsymbol{\Sigma}}(p) = 1.50; 0.60; 0.25; 0.00; -0.15$ for $p = 0, \ldots, 4$. Model: A5.
- 1. Compute the penalty per lag $c_T K^2 / T$ for AIC, BIC and HQ.
- 2. Compute the three criteria for each p and find the minima.
- 3. Say which p you would start from with 80 quarters, and why.
- Report: three penalties, a table of 15 values, three orders and one sentence.

Back Exercise list

B1: does the S&P 500 lead the BET? [Solved]

- **Question**: do yesterday's S&P 500 returns help to forecast today's BET returns, and the other way round?
- **Inputs**: daily log returns (%) of the S&P 500 and the BET since 2000, on common trading days
- **Tasks**
  1. Compute the cross-correlations $\mathrm{Corr}(\mathrm{BET}_t, \mathrm{S\&P}_{t-k})$ for $k = -5, \ldots, 5$.
  2. Choose p for a bivariate VAR with AIC, BIC and HQ (maximum 10) and estimate the VAR with the HQ order.
  3. Test Granger causality in both directions.
  4. Compute the generalised response of the BET to an S&P 500 shock for 5 days.
  5. Interpretation: why does New York lead Bucharest by one day?
- **Report**: three cross-correlations, the chosen p, two F tests, two responses and one sentence
- Notebook: section B1

B1: solution [Solved]



- $T = 6\,466$ days; CCF: 0.21 ($k = 0$), 0.170 ($k = 1$), -0.003 ($k = -1$); orders: AIC 9, BIC 1, HQ 1: VAR(1)
- S&P 500 $\rightarrow$ BET: $F = 156.2$ ($p < 0.001$); BET $\rightarrow$ S&P 500: $F = 2.25$ ($p = 0.133$); coefficient of $\text{S\&P}_{t-1}$: 0.182 (SE 0.015)
- Generalised response of the BET: 0.33% on the same day, 0.24% the next day, -0.007% after two days
- Interpretation: Wall Street closes after Bucharest, so news of the American afternoon reaches the BET only the next day: a time-zone effect, not a slow market

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B2: the same question with weekly returns [Proposed]

- **Question**: does the lead of the S&P 500 survive when we use Friday-to-Friday weekly returns?
- **Inputs**: weekly log returns (%) of the S&P 500 and the BET from the Friday closes; model: B1
- **Tasks**
  1. Repeat steps 1–4 of B1 on weekly returns.
  2. Compare the lag-1 cross-correlation and the F statistic with the daily ones.
  3. Interpretation: why is the lead much weaker at the weekly frequency?
- **Report**: the same items as in B1 and two sentences
- Notebook: section B2

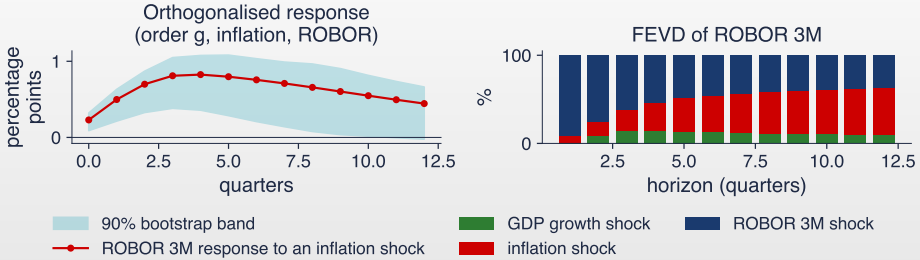
Back Exercise list

B3: a VAR for the Romanian economy [Solved]

- **Question**: how do GDP growth, inflation and the 3-month interest rate interact in Romania?
- **Inputs**: $\mathbf{Y}_t = (g_t, \pi_t, i_t)'$, quarterly, 2005Q1–2026Q2
- **Tasks**
  1. Choose p with AIC, BIC and HQ ($p \leq 6$) and estimate a VAR(2).
  2. Check stability (largest modulus of the companion eigenvalues) and run the portmanteau test with $h = 8$ and $h = 12$.
  3. Run the six Granger F tests.
  4. Plot the orthogonalised response of ROBOR to an inflation shock (order g, π, i) and the FEVD of ROBOR.
  5. Interpretation: do the results show that the central bank reacts to inflation?
- **Report**: three orders, one modulus, two portmanteau tests, a table of six tests, the chart and one sentence
- Notebook: section B3

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B3: solution [Solved]



- Orders: AIC 5, BIC 1, HQ 2; VAR(2) on 84 quarters; largest modulus 0.869: stable; $Q_{\mathbf{8}} = 76.7$ ($\mathbf{p} = 0.023$), $Q_{\mathbf{12}} = 105.3$ ($\mathbf{p} = 0.130$)
- Granger (p): $g \to i$ 0.001, $\pi \to i$ 0.011, $i \to g$ 0.022, $\pi \to g$ 0.059, $i \to \pi$ 0.417, $g \to \pi$ 0.853
- ROBOR after an inflation shock: 0.23 pp on impact, 0.82 pp after 4 quarters; FEVD of ROBOR at 12 quarters: inflation 53%, own 37%
- Interpretation: they show that ROBOR follows inflation, as a reaction function would imply; ROBOR is a market rate, and the VAR cannot separate the BNR's decisions from market expectations

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B4: adding the EUR/RON exchange rate [Proposed]

- **Question**: does the depreciation of the leu help to forecast inflation (exchange-rate pass-through)?
- **Inputs**: $\mathbf{Y}_t = (g_t, s_t, \pi_t, i_t)'$ with s_t the quarterly EUR/RON change, from 2005Q4; model: B3
- **Tasks**
  1. Choose p with the three criteria ($p \leq 4$) and estimate a VAR(2).
  2. Test whether s Granger-causes π, i and g, and whether i and π Granger-cause s.
  3. Compute the share of EUR/RON shocks in the FEVD of inflation at 1, 4, 8 and 12 quarters.
  4. Interpretation: does the result mean that the exchange rate does not affect Romanian prices?
- **Report**: three orders, five F tests, four shares and two sentences
- Notebook: section B4

C1: the price puzzle [Proposed]

- **Question**: why does inflation rise after a ROBOR shock in the VAR of B3, and does it change when the exchange rate or more lags are added?
- **Inputs**: the VARs of B3 and B4; ROBOR ordered last; models: B3, B4
- **Tasks**
  1. Plot the response of inflation to a ROBOR shock in the VAR(2) of B3, the VAR(2) of B4 and a VAR(4) with the four variables of B4.
  2. Report the largest and the smallest response and their horizons.
  3. List two pieces of information that the central bank has and the VAR does not.
  4. Interpretation: is a positive response of inflation a proof that higher rates raise prices?
- **Report**: the chart, six numbers and a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to analyse the VAR of B3. The answer:
- (a) AIC chooses p = 5, so the true lag order of the economy is 5.
- (b) The largest eigenvalue modulus of $\mathbf{A_1}$ is 1.37, so the VAR(2) is explosive.
- (c) The portmanteau test at 8 lags has p = 0.023, so the residuals are white noise.
- (d) GDP growth Granger-causes ROBOR, so faster growth causes the BNR to raise rates.
- (e) Cholesky impulse responses do not depend on the order of the variables.
- (f) In each row of the FEVD the shares add up to 100%.
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- Stability: eigenvalues of $\mathbf{A}$ (or of the companion matrix) inside the unit circle; then forecasts converge to $\boldsymbol{\mu}$
- Granger F test: compare the equation with and without the lags of the candidate variable
- Cholesky responses depend on the ordering; the generalised responses do not
- S&P 500 leads the BET by one day because of the time zones; ROBOR follows Romanian inflation
- An AI answer is a draft: check the direction of each claim and which matrix decides stability

After the seminar

- Details: Chapter 6
  - vector processes, the companion form, lag selection, diagnostics, Granger causality and its pitfalls, impulse responses, FEVD, spillovers, forecasting, structural VARs
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: monetary policy shocks in Romania with an extended VAR
- Reading: Huang and Petukhina (2022), Ch. 7; Hyndman and Athanasopoulos (2021), Sec. 12.3; Stock and Watson (2001)

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